Healthcare Call Center Performance Analysis

Project Overview

This project uses SQL, Power Query, and Power BI to analyze healthcare call center performance across California counties.

The analysis combines three datasets containing call volume, average wait time, and average abandonment rate for January–June 2026 to identify geographic and operational patterns that may indicate areas of service inefficiency. SQL was used to clean and integrate the datasets, validate the resulting data, calculate performance metrics, analyze monthly and county-level trends, and identify counties experiencing elevated operational pressure. Power Query was then used to prepare the data for Power BI, where an interactive dashboard was created to communicate key findings through KPIs and visualizations.

Business Questions

This analysis aims to answer the following questions:

- How many calls were handled across the healthcare call center during the analysis period?
- How did the call volume, wait time, and abandonment rate change month to month?
- Which counties experienced the longest average wait times?
- Which counties had the highest abandonment rates?
- How does the abandonment rate vary across different wait-time levels?
- Which counties experience both high call volume and high abandonment rates?
- Which counties should be considered high priority based on call volume, wait time, and abandonment rate?

- Which counties showed improvement in abandonment rates between January and June?

Key Findings

Overall Performance

- The data set contains 9,562,449 calls across the six-month analysis period
- Average wait-time across county-month observations was approximately 32.87 minutes
- Average abandonment rate was approximately 24.11% (Power BI reports 27% due to differences in aggregation across the three source datasets).

Monthly Performance

- January had the highest call volume at 1,759,226 calls
- May had the lowest call volume at 1,425,221 calls
- Average wait time declined from 41.46 minutes in January to 29.17 minutes in June
- Average abandonment rate declined from 27.76% in January to 22.27% in June

County Performance

- Los Angeles had the highest total call volume at approximately 1.15 million calls
- Sacramento followed with approximately 915,00 calls
- Orange handled approximately 882,000 calls

Wait Time

- Tulare Country had the highest average wait time at 118.41 minutes
- Orange, Santa Barbara, Sacramento, Kern, and Monterey also had average wait times above 60 minutes

Abandonment

- Tulare County had the highest average abandonment rate at 47.17%
- Alameda and Fresno followed with abandonment rates at 45.5% and 44.5% respectively

High-Priority Counties

Using a combined threshold of:

- At least 250,000 calls
- Average wait time of at least 40 minutes
- Average abandonment rate of at least 30%

The analysis identified:

- Tulare
- Los Angeles
- Kern
- Sacramento

as counties experiencing particularly high operational pressure.

Wait Time and Abandonment

The wait-time grouping analysis showed that abandonment increased substantially as wait times increased:

Wait Time	Average Abandonment
Under 15 minutes	6.25%
15-29 minutes	31.03%
30-59 minutes	32.49%
60+ minutes	34.69%

Tools and Technologies:

SQL:

- Data cleaning and validation
- Dataset integration
- Performance, Correlation and Trend Analysis

DB Browser for SQLite:

- SQL development
- Database management
- Analysis result generation

Power Query:

Dataset merging

Preparation of the analytical data set for Power BI

Power BI:

- Dashboard development
- KPI cards
- County Performance Analysis

Data

The analysis uses three healthcare call-center datasets:

Call Volume:

- Reporting Period
- County Name
- Total Call Volume

Average Wait Time:

- Reporting Period
- County Name
- Average Wait Time

Average Abandonment Rate:

- Reporting Period
- County Name
- Average Abandonment Rate

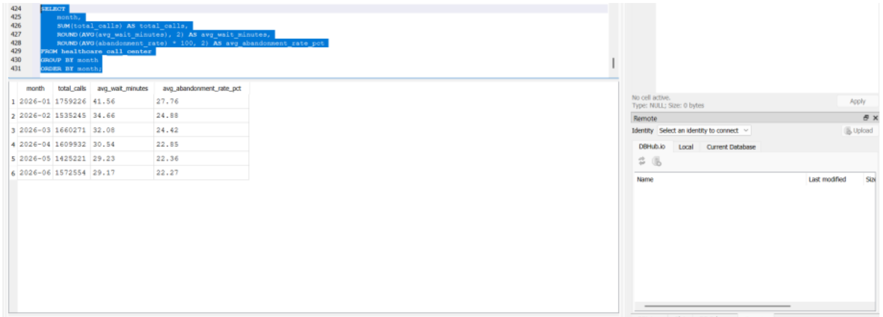
The datasets cover January through June 2026.

SQL Analysis

SQL was used to transform the raw datasets into a single dataset and perform the core analysis.

Key Analyses included:

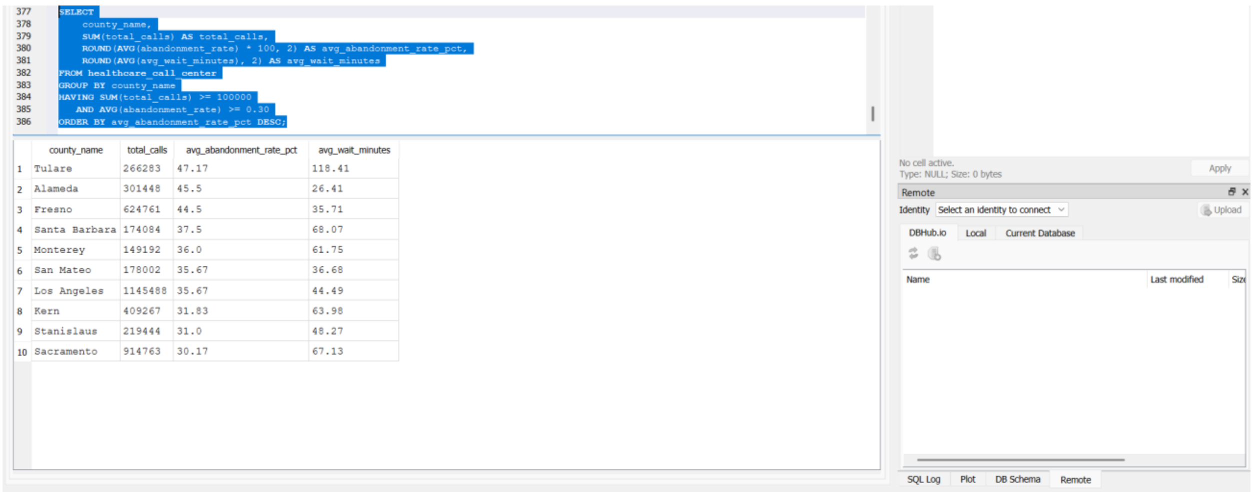
- Monthly performance trends



```sql
SELECT
    month,
    SUM(total_calls) AS total_calls,
    ROUND(AVG(avg_wait_minutes), 2) AS avg_wait_minutes,
    ROUND(AVG(abandonment_rate) * 100, 2) AS avg_abandonment_rate_pct
FROM healthcare_call_center
GROUP BY month
ORDER BY month;
```

	month	total_calls	avg_wait_minutes	avg_abandonment_rate_pct
1	2026-01	1759226	41.56	27.76
2	2026-02	1535245	34.66	24.88
3	2026-03	1660271	32.08	24.42
4	2026-04	1609932	30.54	22.85
5	2026-05	1425221	29.23	22.36
6	2026-06	1572554	29.17	22.27

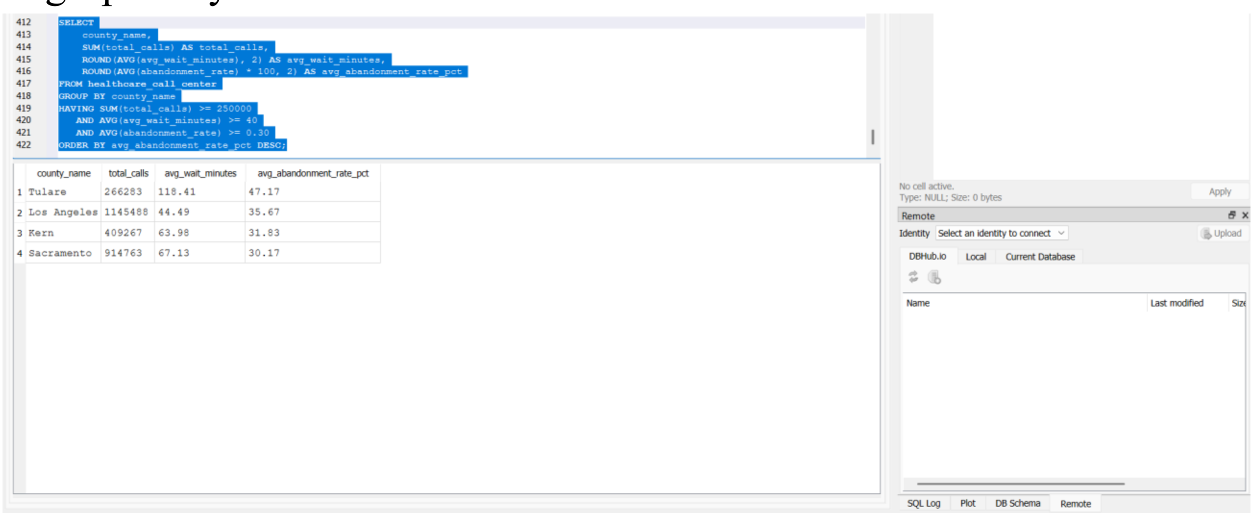
- High-volume counties with a high abandonment rate



```sql
SELECT
    county_name,
    SUM(total_calls) AS total_calls,
    ROUND(AVG(abandonment_rate) * 100, 2) AS avg_abandonment_rate_pct,
    ROUND(AVG(avg_wait_minutes), 2) AS avg_wait_minutes
FROM healthcare_call_center
GROUP BY county_name
HAVING SUM(total_calls) >= 100000
   AND AVG(abandonment_rate) >= 0.30
ORDER BY avg_abandonment_rate_pct DESC;
```

	county_name	total_calls	avg_abandonment_rate_pct	avg_wait_minutes
1	Tulare	266283	47.17	118.41
2	Alameda	301448	45.5	26.41
3	Fresno	624761	44.5	35.71
4	Santa Barbara	174084	37.5	68.07
5	Monterey	149192	36.0	61.75
6	San Mateo	178002	35.67	36.68
7	Los Angeles	1145488	35.67	44.49
8	Kern	409267	31.83	63.98
9	Stanislaus	219444	31.0	48.27
10	Sacramento	914763	30.17	67.13

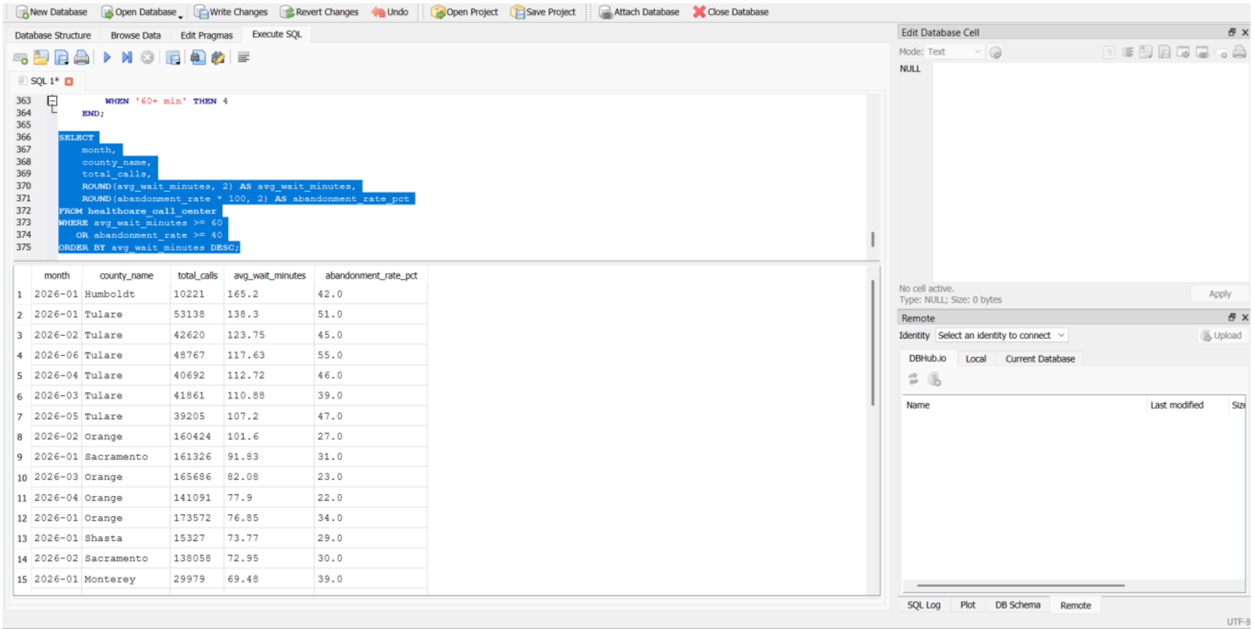
- High-priority counties



```sql
SELECT
    county_name,
    SUM(total_calls) AS total_calls,
    ROUND(AVG(avg_wait_minutes), 2) AS avg_wait_minutes,
    ROUND(AVG(abandonment_rate) * 100, 2) AS avg_abandonment_rate_pct
FROM healthcare_call_center
GROUP BY county_name
HAVING SUM(total_calls) >= 250000
   AND AVG(avg_wait_minutes) >= 40
   AND AVG(abandonment_rate) >= 0.30
ORDER BY avg_abandonment_rate_pct DESC;
```

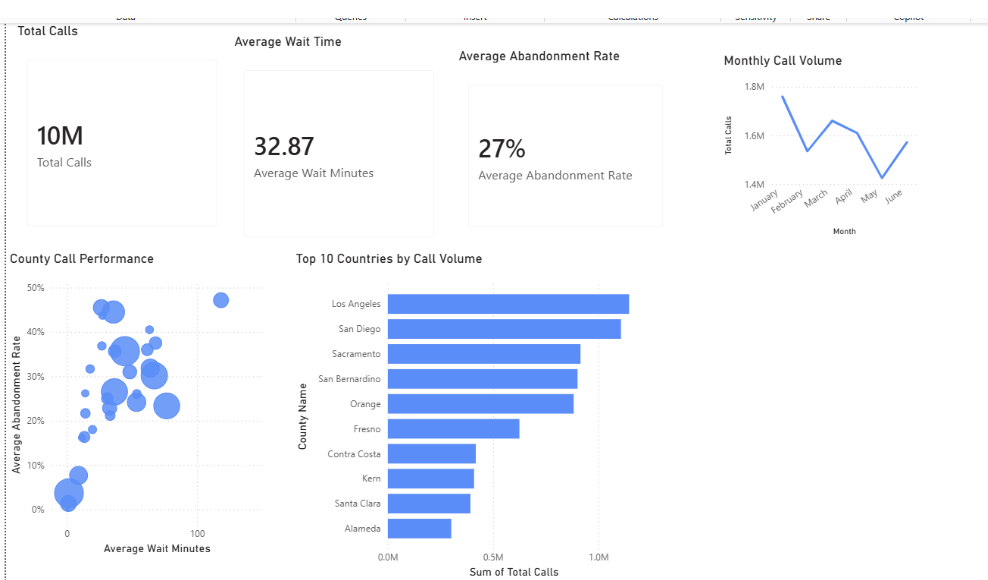
	county_name	total_calls	avg_wait_minutes	avg_abandonment_rate_pct
1	Tulare	266283	118.41	47.17
2	Los Angeles	1145488	44.49	35.67
3	Kern	409267	63.98	31.83
4	Sacramento	914763	67.13	30.17

- Wait-time and abandonment patterns



Power BI Dashboard

The Power BI dashboard transforms the SQL analysis into an interactive report designed to provide an executive overview of healthcare call-center performance.



Executive Dashboard

The dashboard includes:

- Total Calls
- Average Wait Time
- Average Abandonment Rate

Visualizations:

- Monthly Call Volume
- County Call Performance
- Top 10 Counties by Call Volume

Business Insights

The SQL analysis indicates that call-center performance improved during the six-month period. While January had the highest call volume and the highest average wait time, both average wait time and abandonment rate declined by June.

However, performance varied considerably across counties.

Some counties experienced a combination of high call volume, long wait times, and high abandonment rates, indicating potential areas for operational attention.

Tulare represents the clearest example of this pattern, with 266,283 calls, an average wait time of 118.41 minutes, and a 47.17% abandonment rate.

The wait-time analysis also shows that abandonment was substantially higher among county-month observations with longer wait times, suggesting that reducing excessive wait times may be an important opportunity for improving caller retention.

Project Files

Healthcare Call Center Analysis.sql: SQL cleaning, integration, KPI, and performance analysis queries
Total Call Volume.csv : Call-volume source data
Average Wait Time.csv: Wait-time source data
Average Abandonment Rate.csv: Abandonment-rate source data
Healthcare Call Center Dashboard.pbix : Interactive Power BI report
Power Query Transformation: Data preparation and dataset merging
SQL Overall KPIs.png: Overall performance analysis
SQL Monthly Performance Analysis.png: Monthly performance results
SQL Monthly Performance Trends.png: Monthly trend analysis
SQL High Volume County Performance.png: High-volume County analysis
SQL High Volume High Abandonment.png: High-volume/high-abandonment analysis
SQL High Priority Counties.png: High-priority county analysis
SQL Wait Time Abandonment Analysis.png: Wait-time/abandonment analysis
SQL High Wait AbandonmentOutliers.png: High-wait/high-abandonment observations